

SAS — Milestone & Template Creation Guide

Shipment Alerting System · How to build milestones, assemble templates, and how the system works

1. How the system works

Every alert in SAS follows the same pipeline. You never send an email by hand — you describe **what to watch** and **who to tell**, and the system does the rest.

- **Build a milestone** — one thing to watch on a shipment (a date, a missing field, a comparison, or a document) together with its alert rules.
- **Add milestones to a template** — an ordered sequence for a shipment type (e.g. “Air Export”).
- **Assign the template to shipments** — this freezes a copy of each milestone and its rules onto every matching shipment, so later edits don’t disturb shipments already in flight.
- **The alert engine** reads those frozen rules on a schedule and sends reminders, stopping automatically when the condition clears.

Note: the runtime alert engine is not wired up yet. Everything you configure is saved correctly and staged — emails begin firing once that engine is added.

You can build a milestone in two places, both using the same 4-step builder: in the **Milestone Library** (reusable across templates) or **inline while building a template** (kept to that template unless you tick “Also save to library”).

2. The 4-step milestone builder

Step 1 — Basic Info

Field	What it means	Tip
Milestone Name	The label shown everywhere (e.g. “Cargo Ready”).	Name it after the event, not the field.
Milestone Type	What kind of check this is (four types below). Decides what Step 2 asks for.	Pick this first; it changes the form.
Priority (Critical)	Toggle. Critical milestones show in red and are meant for escalation.	Use for live / time-sensitive cargo.
Description	Optional note on what it tracks and why.	Helps teammates reusing it later.

The four milestone types:

- **Date Check** — watches a date field; alerts before / on / after it. (Cargo Ready, Pickup, Arrival, Delivery.)
- **Missing Info** — alerts while a required field is empty; stops when filled. (Consignee email, contact, carrier.)
- **Field Comparison** — alerts when a value meets a condition vs another field or a fixed value. (Delay detection, date mismatch.)
- **Document Check** — alerts when a document hasn’t been updated, tracked via a “last edited” timestamp. (Bill of Lading.)

Step 2 — Field Linking

Connects the milestone to real shipment data; what you see depends on the type. Fields that aren't valid for the current type appear greyed out.

Date Check / Missing Info

Field	What it means
Primary Field	The field this milestone watches (e.g. Cargo Pickup Date). For Missing Info, the field that must not stay empty.
Expected Date Source (Date)	How the due date is worked out: use the field's own value · calculate from another field (+/- N days) · N days after shipment creation · set manually at assignment.

Field Comparison — build a sentence: [Field A] [operator] [Field B or fixed value]

Field	What it means
Field A	The field being tested (Pickup Status, Delay Days).
Operator	Only operators valid for Field A's type are selectable (dates, numbers, text each get their own).
Field B / Fixed value	Compare to another field of the same type, or type a fixed value (Delayed, 3).
Threshold	The number when the operator needs one ("more than 3 days").

Document Check

Field	What it means
Document Name	The document's name, shown in the alert email ("Bill of Lading").
Tracking Field	The timestamp that changes when the document is updated — usually a "last edited" field. Date fields only.

Step 3 — Alert Rules

A milestone can have several rules; each fires independently. For every rule:

When to fire — Before / On date / After the due date, N days offset, at a time (e.g. 09:00).

Fire condition — the alert only sends if this is true at fire time:

- **Always** — send no matter what.
- **If not recorded** — send only if the primary date is still empty. This is what makes date reminders switch off once the date arrives.
- **If condition** — send only if the comparison is still true (comparison milestones).
- **If missing** — send only if the required field is still empty (missing-info milestones).

Recurrence — Once · Daily · Weekly · Custom (every N days).

Stop repeating when (for anything other than Once):

- **After N times** — stop after N sends.
- **On date** — stop on a calendar date.
- **Condition met** — pick a Watch field and how it ends: Has a value / Is empty / Equals value / Changes. Use this to auto-stop when the data updates.
- **Never (manual)** — only stops when cancelled by a person.

Who receives this alert — Operations (created_by_email) · Sales (sales_user_email) · Consignee (consignee_email) · Custom (an email you type).

The Operations / Sales / Consignee addresses come from the shipment record. If those email columns aren't populated by the CargoWise sync yet, only Custom will have somewhere to send until that data is filled in.

Step 4 — Preview & Save

- **Save to template only** — stays local to this template (inline build).
- **Also save to library** — saved to the library and added to the template (tick the box in Build New).
- **Save to library** — from the library page, it just goes to the library, ready to reuse.

3. Building a template

- Go to Milestones → Templates List → New Template.
- Give it a name and shipment type, optional description.
- Add milestones in order: **Pick from Library** (comes with its rules) or **Build New Milestone** (full builder inline).
- Reorder with the up/down arrows; remove with the ×.
- Save Template, then Assign to Shipments (All / by type / custom). Assignment snapshots each milestone + its rules onto the shipments.
- To make a template-only milestone reusable later, open the template and click **Add to Library** on it.

4. Worked examples (recipes)

A. “Cargo Ready” reminder that stops once recorded

- Type: Date Check · Primary field: Cargo Ready Date · Due date: N days after creation (or manual).
- Rule: Before 1 day at 09:00 · If not recorded · Daily · Stop when Condition met → Cargo Ready Date → Has a value · Operations.
- Result: nags daily until the date is filled, then stops on its own.

B. “Pickup delayed” escalation

- Type: Field Comparison · Field A: Pickup Status · Operator: equals · Fixed value: Delayed.
- Rule: On date at 08:00 · If condition · Daily · Stop when Pickup Status → Changes · Operations (add a second rule to Sales after 2 days).

C. “Consignee email missing” data-quality check

- Type: Missing Info · Primary field: Consignee Email.
- Rule: After 0 days at 10:00 · If missing · Daily · Stop when Consignee Email → Has a value · Operations.

D. “Bill of Lading not uploaded”

- Type: Document Check · Document Name: Bill of Lading · Tracking field: Documents Last Edited.
- Rule: Before 2 days at 09:00 · If not recorded · Daily · Stop when Documents Last Edited → Changes · Operations.

Starter “Air Export” template (5 milestones in order): Cargo Ready → Cargo Pickup (+ delayed comparison) → Departure → Arrival → Delivery / Customer Handover — all Date Checks on their matching date fields.

5. Quick rules of thumb

- **“Remind until it’s done”** → If not recorded (dates) or If missing (fields) · Daily · Stop when Has a value.

- **“Alert only while a problem is true”** → Field Comparison + If condition · Stop when Changes.
- **Critical shipments** → turn on Critical and add a second rule escalating to Sales after a day or two.
- **Reuse anything you’ll need again** → save it to the Library so future templates can just pick it.